

ECEN 4517 - PV Power Lab
Experiment 5 Report
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2 Modified Sine-Wave Inverter Circuitry

2.1 Circuit Board Construction

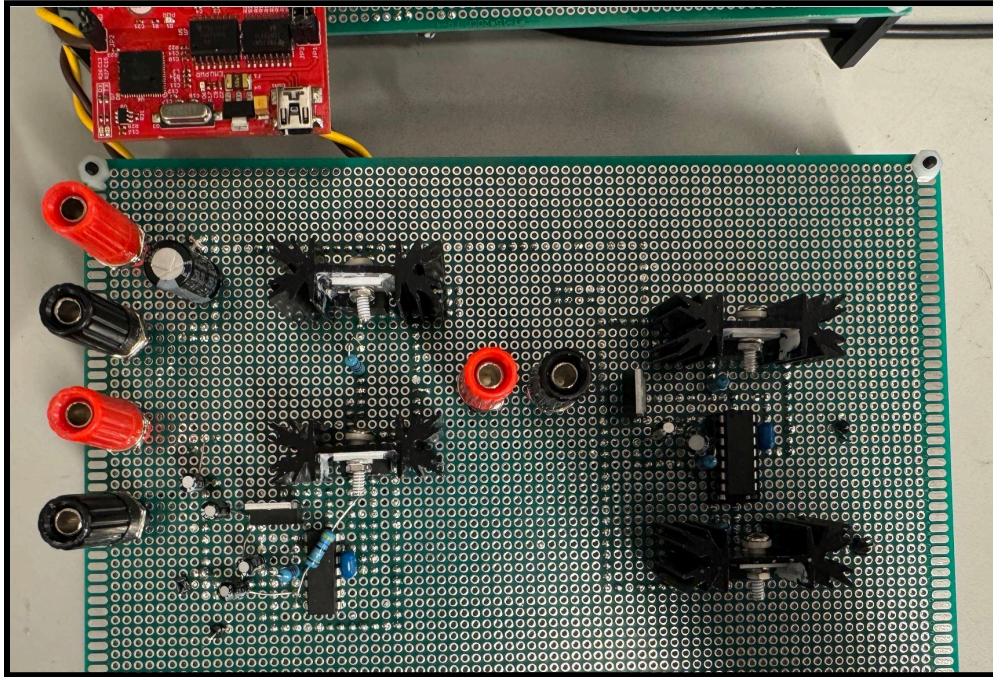


Figure 2.1 - 1: Top View of the Inverter

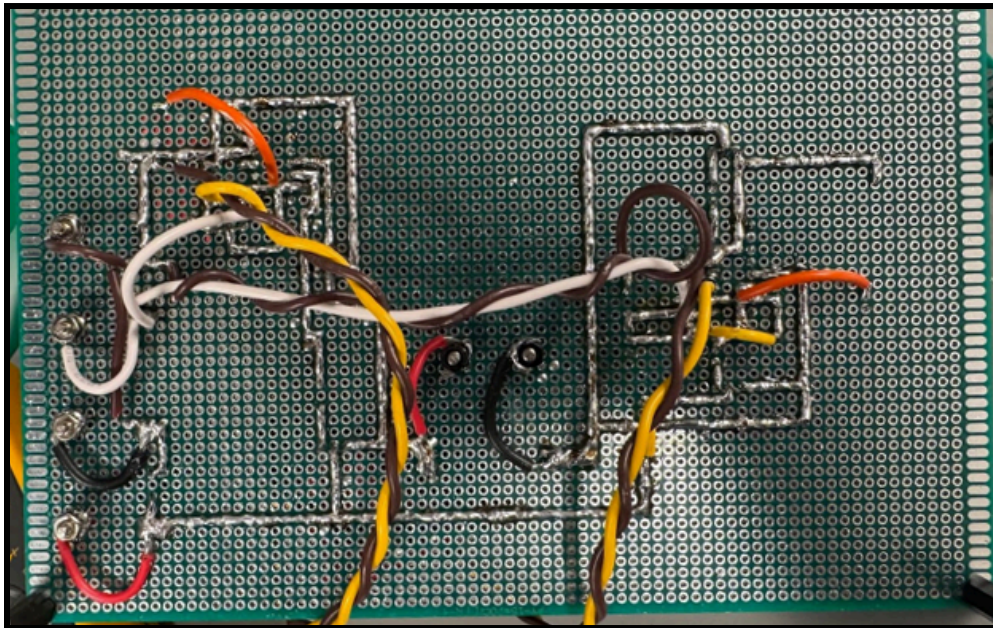


Figure 2.1 - 2: Bottom view of the Inverter

The above Figures, 2.1 - 1 and 2, show the fully constructed modified sin wave inverter circuit.

3 Modified Sine-Wave Gate Control Signals

3.1 Gate Driver Controller Code

To create a modified sin wave, the half_duty variable was adjusted to 20,000 and the CTRMODE bit field was set to TB_COUNT_UPDOWN. These changes adjusted the duty cycle and phase of the gate driver waveforms. Other changes to the code are shown below.

```
// Setup counter mode
EPwm3Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; //CHANGED TO UPDOWN // This sets the PWM type, ie trailing, leading or symmetric
EPwm3Regs.TBCTL.bit.PHSEN = TB_ENABLE; // Enable phase loading
EPwm3Regs.TBCTL.bit.HSPCLKDIV = 0; // Clock ratio to SYSCLKOUT
EPwm3Regs.TBCTL.bit.CLKDIV = 3; // PWM clock prescale set to 8. Timebase clock frequency = (

// Setup shadowing
EPwm3Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm3Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm3Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // Load on Zero
EPwm3Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO;

// Set actions
EPwm3Regs.AQCTLA.bit.CAD = AQ_SET; //CHANGED TO CAD // Sets PWM1A J6.5 when period starts
EPwm3Regs.AQCTLA.bit.CAU = AQ_CLEAR; // Clears PWM1A J6.5 when CMPA > sawtooth
```

```
// Setup counter mode
EPwm4Regs.TBCTL.bit.CTRMODE = TB_COUNT_UPDOWN; //CHANGED TO UPDOWN // This sets the P
EPwm4Regs.TBCTL.bit.PHSEN = TB_ENABLE; // Enable phase loading sets the phas
EPwm4Regs.TBCTL.bit.HSPCLKDIV = 0; // Clock ratio to SYSCLKOUT
EPwm4Regs.TBCTL.bit.CLKDIV = 3; // PWM clock prescale set to 8. Timeb

// Setup shadowing
EPwm4Regs.CMPCTL.bit.SHDWAMODE = CC_SHADOW;
EPwm4Regs.CMPCTL.bit.SHDWBMODE = CC_SHADOW;
EPwm4Regs.CMPCTL.bit.LOADAMODE = CC_CTR_ZERO; // Load on Zero
EPwm4Regs.CMPCTL.bit.LOADBMODE = CC_CTR_ZERO;

// Set actions
EPwm4Regs.AQCTLA.bit.CAD = AQ_SET; //CHANGED TO CAD // Sets PWM1A J2.8 when period starts
EPwm4Regs.AQCTLA.bit.CAU = AQ_CLEAR; // Clears PWM1A J2.8 when CMPA > sawtooth
```

Figure 3.1 - 1: Gate Driver Control Code

4 Modified Sine-Wave Inverter Testing

4.1 Voltage and Current Waveforms

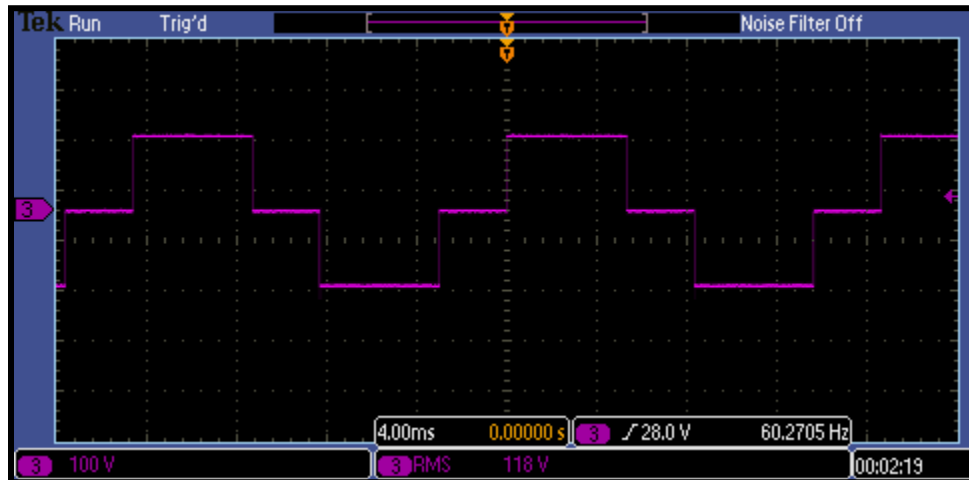


Figure 4.1 - 1: Output Rms Voltage Waveform

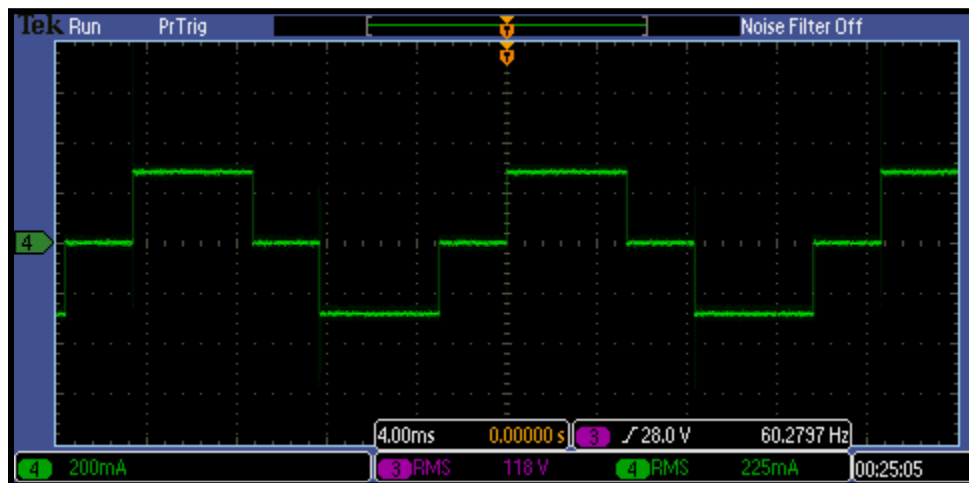


Figure 4.1 - 2: Output Rms Current Waveform

Above are the output rms voltage and current waveforms for the modified sine wave inverter. The output inverter was able to produce 119.6 Vrms and 0.287 Arms. This modified sin wave ran at 60.28 Hz representing a common household wall outlet.

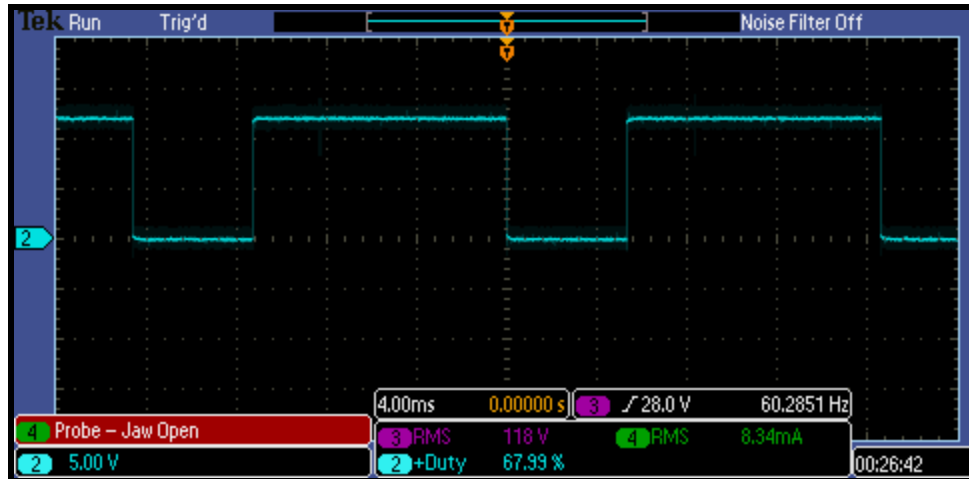


Figure 4.1 - 3: Gate Driver Output Voltage Waveform

To find our duty cycle, a measurement of the gate driver was taken. The duty cycle was measured to be around 68%.



Figure 4.1 - 4: Output Rms Voltage and Current.

Figure 4.1 - 4 shows our measured input and output voltage and current . The top multimeters show the measured DC input voltage and current, while the bottom two multimeters show the output AC voltage and current.